

FITTING SHOP

PRACTICE-I

Time : 3 Hrs.

OBJECT:- To prepare a practice work piece as per given drawing involving measuring, marking & filing operation.

MATERIAL REQUIRED :- M.S. Flat 50mm x 50mm x 10mm.

TOOLS AND EQUIPMENTS USED:-

1) Measuring and Marking Tools:

- a) Surface Plate
- b) Angle Plate
- c) Universal Surface Gauge
- d) Try Square
- e) Engineering Scale
- f) Centre Punch
- g) Calipers

2) Holding and Supporting Tools:

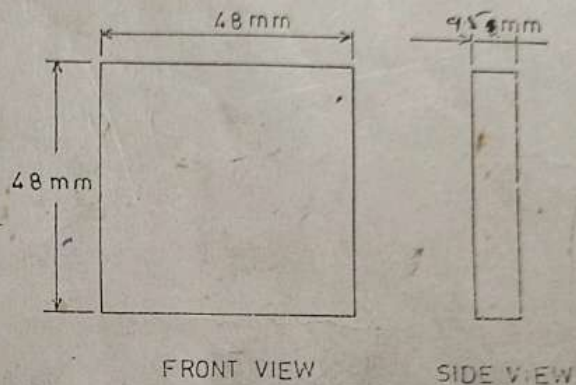
- a) Bench Vice

3) Cutting Tools:

- a) File, (b) chisel

4) Striking Tools:

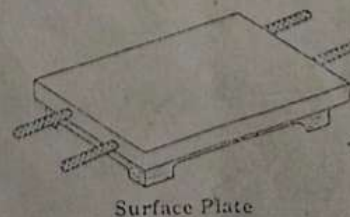
- a) Ball Peen Hammer



DESCRIPTION OF TOOLS AND EQUIPMENTS :-

1. a) Surface Plate :

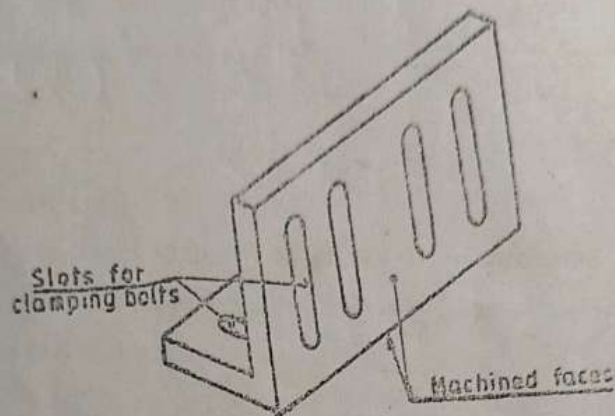
Surface plate is used for testing the flatness, trueness of the surface. It is made of gray cast iron. Its upper face is planed to form a very smooth surface. It is covered with a wooden cover while not in use.



Surface Plate

b) Angle Plate :

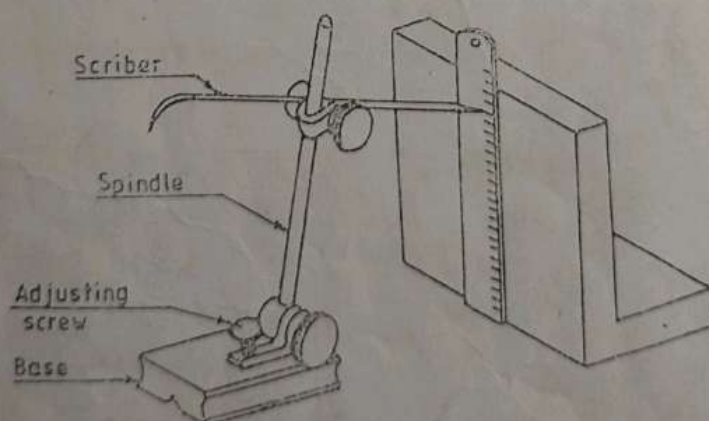
It is made of cast iron in different sizes. It has two planed surface at right angle to each other. It has various slots in both the surfaces to hold the work firmly by means of bolts and clamps. It is used for holding & supporting the job.



Angle plate

c) Universal Surface Gauge :

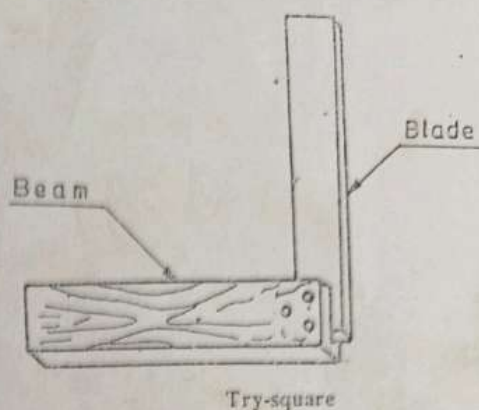
It consists of a heavy base, a scriber and a bar. The scriber can be adjusted to any position with the help of screw and nut. It is used for scribing parallel lines.



Universal Surface Gauge

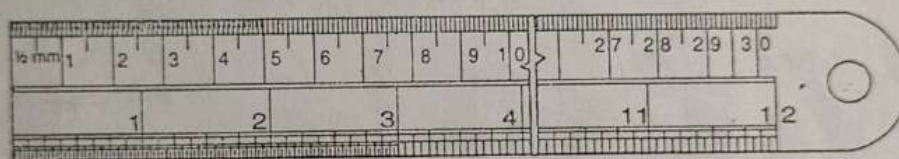
d) **Try Square :**

It is used for checking squareness of surfaces and set off lines at right angle to given edge. It consists of a 'blade' made up to steel which is attached to a base at 90° checking right angles and measuring straightness of surface.



e) **Engineering Scale :**

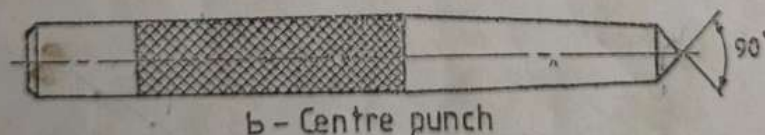
It is a stiff straight steel strip having all the faces machined true on one of the flat faces, graduations are marked in inches and centimeters. The least count is $1/64$ inch & 0.5 mm.



Engineering Scale

f) **Centre Punch :**

It is primarily a marking tool used in bench work. It is generally made from octagonal cast steel about 100mm long. It is used to mark the centre of the hole before drilling. One end of punch carries point ground to an incline angle 90° . The body is knurled for gripping by hand.



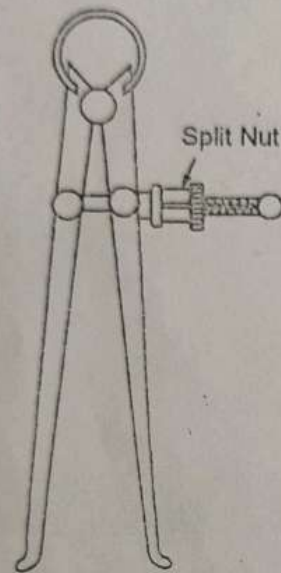
b - Centre punch

g) **Calipers :**

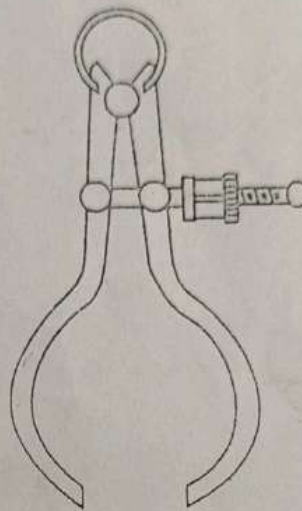
It is generally used to measure the inside or outside diameters & lengths. It is made in different sizes and shapes. It consists of two bent legs connected at one end by means of rivets or bolts. The edges are made to

just touch the job. Then the dimension is measured with the help of steel scale. There are three types of calipers.

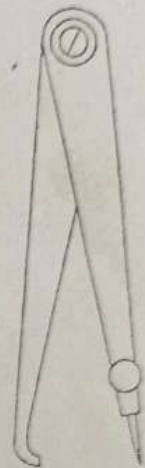
- (a) Inside caliper
- (b) Outside caliper
- (c) Hermaphrodite caliper



(ii) Inside Type



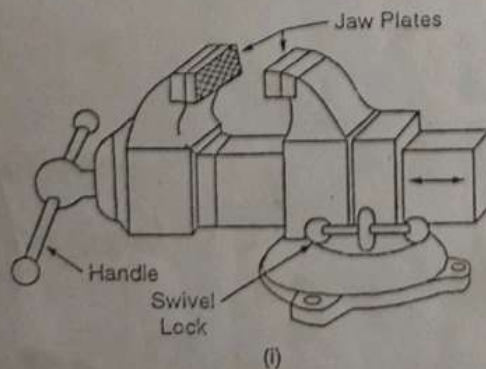
(i) Outside Type



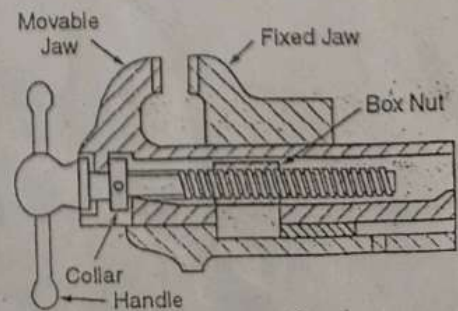
Hermaphrodite

2. a) Bench Vice :

Vice is used for gripping and holding the work in position during various operations such as filing, chipping & scraping. Bench vice fixed on the bench with the help of nut & bolts. It consist of cast iron body fixed jaw, movable jaw both of cast iron, spindle, handle, jaw plates, nut, spring and washer. The height of vice should be such that the vice jaws are about same height as the operators elbow. The size of the vice is given by width of the jaws and maximum opening between the jaws.



(i)

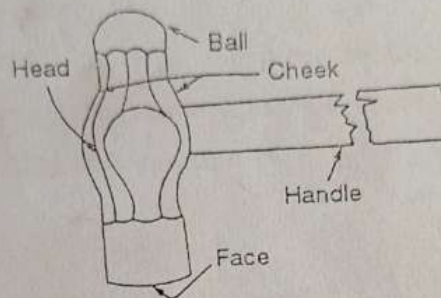


(ii)

Bench Vice

3. a) Ball Peen Hammer :

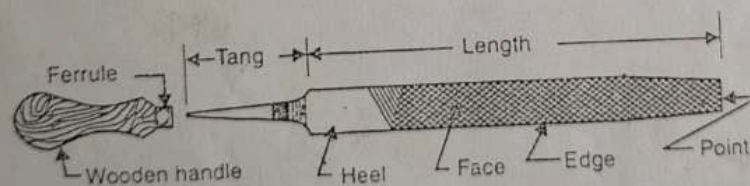
This hammer is most commonly used. It has round ball shaped head and is used for all general purpose work. The peen and face are hardened. The middle portion of the hammer is not hardened so as to absorb the shocks. It is mostly used for riveting i.e. forming a shape head and a tail of rivet and chipping.



1 Ball peen hammer

4. a) File :

A file is a hardened piece of high carbon steel with steel with slanting rows of teeth. It is used to cut, smooth or fit metal parts in each other. It cuts all metals except hardened steel and it cut only on the forward stroke. It consists of a blade or body with a tang for fixing into the wooden handle. The teeth are cut on the blade which are hardened & Tempered. The metal ring on the file handle is called ferrule provided in order to prevent splitting of the handle.



Parts of a file.

PROCEDURE :-

1. Check the given piece of raw material and ensure that it is not under size.
2. Hold it in vice and file one flat face. Check its trueness with the help of try square.

3. Hold it in vice again and file an adjacent side surface. Check that these two surfaces are at right angle with the help of try square.
4. Now, apply wet chalk on the filed flat face and let it get dried.
5. Mark overall dimensions and counters (curvature at one corner and angular cut at the other). Make permanent marks along this lines by means of hammer and centre punch.
6. File all the four sides to bring the piece to the required size. (48mm x 48mm) keep checking the squareness of these side surface with their adjacent surface during the operation.

PRECAUTIONS :-

1. Job should be held tightly in vice in right position.
2. Filing should be done very carefully and pressure should be applied only in forward strokes.
3. Don't do hammering on surface plate.

PRACTICE-II

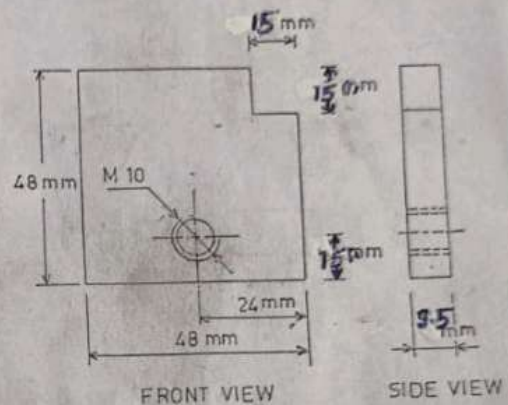
Time : 3 Hrs.

OBJECT:- To prepare a practice work piece as per given drawing involving measuring, marking, sawing, filing, drilling & tapping operation.

MATERIAL REQUIRED :- M.S. Flat 50mm x 50mm x 10mm.

TOOLS AND EQUIPMENTS USED:-

- 1) **Measuring and Marking Tools:**
 - a) Surface Plate
 - b) Angle Plate
 - c) Universal Surface Gauge
 - d) Try Square
 - e) Engineering Scale
 - f) Centre Punch
 - g) Calipers
- 2) **Holding and Supporting Tools:**
 - a) Bench Vice
 - b) Drill Chuck
 - c) Machine Vice
 - d) Tap Handle or Tap Wrench
- 3) **Cutting Tools:**
 - a) File
 - b) Hacksaw
 - c) Drill
 - d) Hand Tap Set
- 4) **Striking Tools:**
 - a) Ball Peen Hammer
- 5) **Machine:**
 - a) Pillar Type Drilling Machine

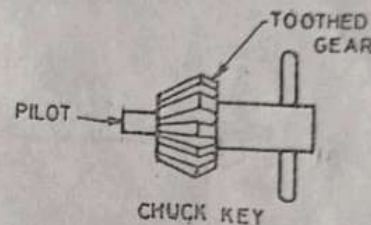
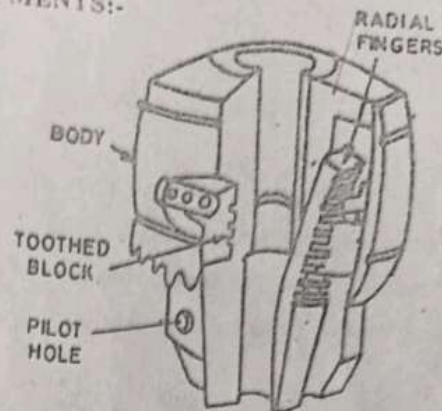


DESCRIPTION OF TOOLS AND EQUIPMENTS:-

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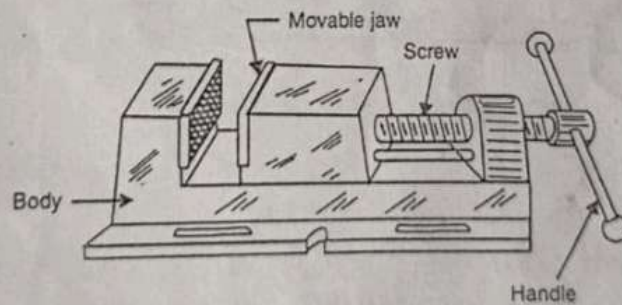
2. b) Drill Chuck :

A drill chuck is the most popular and useful device for holding parallel shank drill and other small tools. A drill chuck is capable of holding a fairly wide range of drill size and it is usually provided with radial fingers for this purpose. The chuck key is used for rotating the toothed body of chuck.



c) Machine Vice :

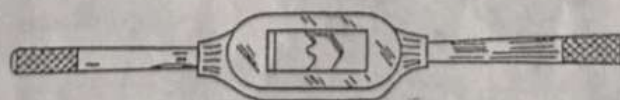
It is particularly used for holding small work for filing or drilling. It is made of mild steel. It consists of a body with a solid jaw, a movable jaw, a screw and a handle for the control of movement of movable jaw.



Machine Vice

d) Tap handle or tap wrench :

It is used for tap holding purpose. It consists of handle, body, fixed jaw & moveable jaw.



Tap wrench

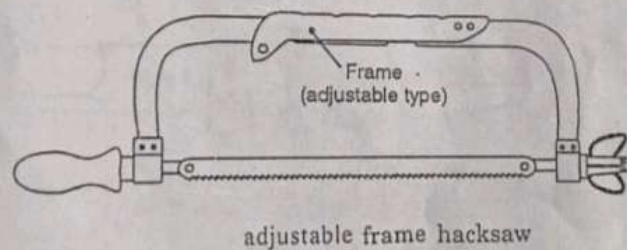
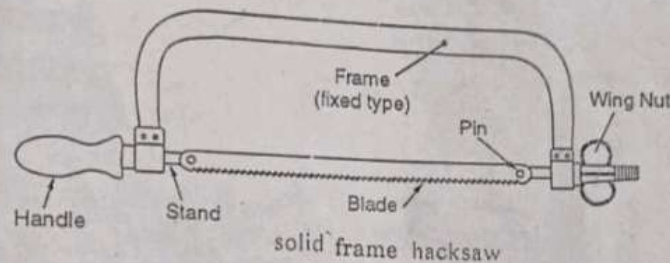
4. b) Hacksaw :

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It is used cutting rods, bars and pipes into desired length. It consists of a metal frame, a wing nut and a blade. Hacksaw consists of two major part.

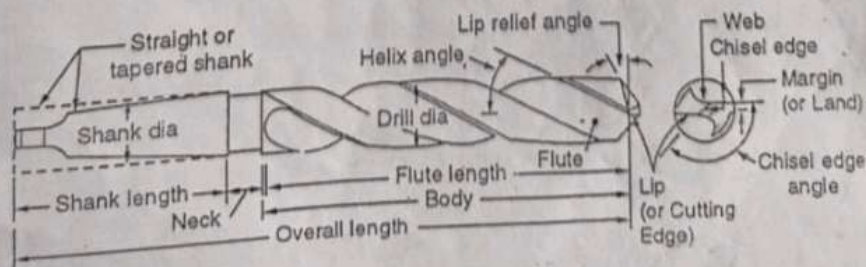
- 1) The frame
- 2) Blade

The frame may be solid or adjustable as shown in figure. The fixed type accommodates the same length of blade always. While adjustable can accommodate blade with different length. The cutting blade has holes at both ends to fit with the hacksaw frame. Blades are made of hard and tempered alloy steel with set teeth.



c) Drill:

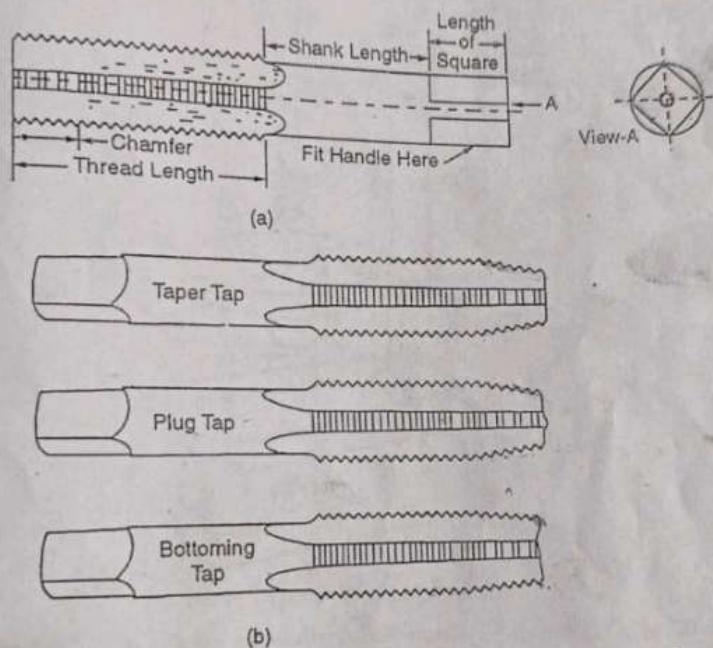
The tool used for making round holes is called drill. It is made up of high speed steel. Containing 18% tungsten, 4% chromium and 1% vanadium. Twist drill is commonly used variety of drill and is made with two, three or four cutting lips. It has cylindrical body as which grooves are cut. These grooves are called flutes. These flutes carry the chips during drilling operation.



d) Hand Tap Set:

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Taps are used for making internal threads into hole. It is inserted into the face of the hole at right angle. In hand tapping a set of three taps is used. Taps are made from high carbon steel or high speed steel. The shank of the tap is left plain and end is squared to accommodate the tap wrench.

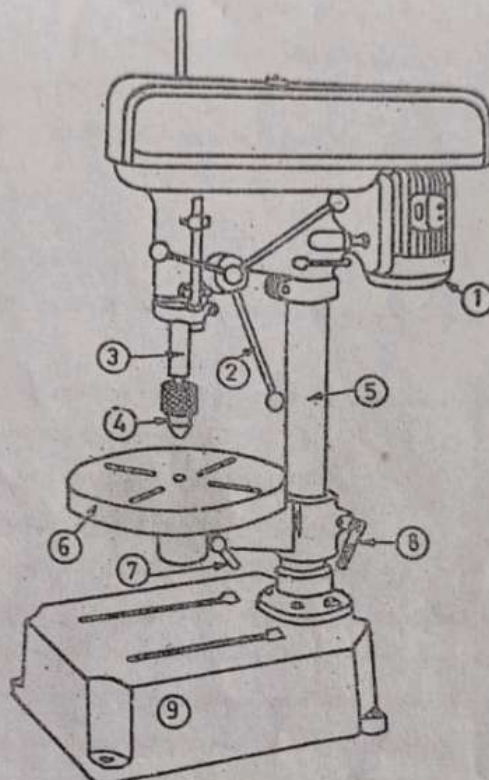


Parts of a tap are shown at (a). The three taps (available in a set) for tapping holes are shown at (b)

5. a) Pillar Type Drilling Machine :

A very commonly used drilling machine, used for light and precision work and normally available in a training workshop. The main parts include a column, base, swiveling table, table clamp, spindle, vertical feed handle and spindle drive mechanism at the top with electrical motor. The cutting tool i.e. drill bit is held in the drill chuck and vertical movements given to it by means of the feed handle to feed

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the revolving tool into the work piece, held on the machine table, to drill the required hole.



Pillar Type Drilling Machine

1. Motor. 2. Feed handle. 3. Spindle. 4. Drill chuck. 5. Column. 6. Table. 7. Table Swivel clamp. 8. Table clamp. 9. Base.

PROCEDURE :-

1. Check the dimension of given piece and ensure that it is not undersize.
2. Apply wet chalk on the filed flat face and let it get dried.
3. Mark over-all dimension and make permanent marks along these line by means of hammer and centre punch.
4. Locate the centre of the hole to be tapped and mark permanently by using hammer & centre punch.
5. Remove the extra material from the jobs near the marking by hacksaw leaving a little margin for filing.